

Generative AI for Instructional Design: Changes, Chances, Challenges

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Generative AI has thoroughly permeated the work of instructional designers. It can be used for a wide variety of tasks such as creating a course map, scripting a case study, drafting handouts, creating visualizations, evaluating alternatives, producing audiovisual media, supporting digital accessibility, checking alignment, creating documentation, preparing slide decks. Generative AI is not a single “magic button” that completely replaces instructional design tasks, but often offers modular capabilities that slot into what instructional designers already do.

Adoption does not come without risks and trepidations: AI can nudge which options feel reasonable, and which paths seem worth pursuing—subtly shaping design choices through defaults, framing, and suggested next steps. Similarly, safety filters in generative AI are not just blocking content; they often work by rerouting prompts into safer cultural templates. Over time, this can result in making materials more uniform and less interesting. Already there is a general wariness of “AI-slop”, verbose content that is superficial, stereotypical or uncontroversial to the point of meaningless. Using generative AI for tasks that were previously done manually may over time erode creative agency: When designers routinely hand tasks to AI, it can result in over-reliance (cognitive offloading) and dependence.

Bias persists, and in some cases may be amplified as models increasingly train on synthetic data. Bias is particularly noticeable in visualizations, where meme culture and stock-photo associations offer dataset priors. Even if users add prompt for diverse depictions, weights may be too low to override defaults, and safety layers may block or dilute sensitive demographic constraints in contexts that are flagged as high-risk. Inaccuracies and hallucinations haven't disappeared completely, which means plausible-looking output can still be wrong in ways that are hard to detect.

These concerns coexist with unresolved questions about data privacy and copyright. Recent disputes around Reddit data illustrate a shift from copyright toward licensing: Reddit has entered paid data-licensing partnerships with OpenAI and Google while it has sued Anthropic alleging that Anthropic scraped and used Reddit content to train models without a licensing agreement. The takeaway for social media users is that selling data for AI training becomes a key revenue source for platforms. Some academic publishers are doing something quite similar. For example, Wiley has reported AI content licensing as a meaningful revenue source. This complicates the relationship between generative AI products and instructional design in academic contexts especially in the open content and open access movements. As an example, [danah boyd](#) recently proclaimed: *"I am a book author and I post my work freely online to ensure that those without resources can read it. But that doesn't mean I want my work taken without permission to train systems for others' profit. That's called exploitation."*

Professional Adoption

On today's job market, instructional designers aren't just competing with other professionals; they're being evaluated against AI-enabled performance, and it's increasingly plausible that employers will soon ask, "Why hire this person instead of relying on AI for this work?". At the same time, the baseline has shifted dramatically. As Helen Crompton pointed out: An average person now has access to more computing power than a Fortune 500 company did 30 years ago (Crompton, 2025)—putting capabilities that once required institutional infrastructure into the hands of individuals with a laptop and a login.

To most instructional designers the central question is no longer whether to use generative AI, but how to apply it in line with professional standards. Early research studies have covered the viewpoints of practitioners:

- The multiple-case study by Kozan et al. (2025) examined six professional instructional designers' integration of GenAI into their professional practice and the factors affecting this integration through semi-structured interviews. The researchers found that instructional designers mostly integrate GenAI into instructional design and development phases, where they believe it has the largest impact. The study revealed that designers' integration of GenAI is mainly based on ambivalent attitudes toward it, closely linked to the advantages and disadvantages associated with the technology, particularly valuing its efficiency in handling tedious tasks and brainstorming while expressing concerns about quality and reliability.
- Luo et. al (2025) surveyed 70 instructional designers and interviewed 13 of them to understand their perceptions and experiences utilizing GenAI across a spectrum of ID tasks. The survey results indicated IDs' familiarity with and perceived usability of GenAI tools in performing various ID responsibilities in their specific contexts. Qualitative findings indicated primary use categories as (1) brainstorming ideas, (2) handling low-stake tasks, (3) streamlining design process, and (4) enhancing collaborations. The study highlighted both the potential for increased efficiency and concerns about content quality, data security, and ethical implications.
- McNeill et al. (2025) surveyed 144 instructional designers on current adoption, tasks, benefits, and concerns regarding generative AI integration. Analysis revealed widespread mainstream usage with 83% leveraging ChatGPT. Accelerating efficiency ranked as the top benefit, with 67% achieving moderate-to-significant time savings that allow more strategic work. Additional gains centered on accelerated content drafting, feedback, and ideation. However, key challenges included verifying accuracy, addressing ethical risks, formulating effective prompts, and lacking personalization. The study concluded that while meaningful automation freed up capacity, truly customized innovation still requires human oversight.
- Yang & Stefaniak (2025) employed Q methodology to explore 19 practicing instructional designers' perceptions of integrating ChatGPT in their design practices. Findings revealed three distinct types of factors: (1) Pessimistic Evaluators, (2) Optimistic Advocates, and (3) Wary Thinkers. The study revealed that instructional designers mainly used ChatGPT to generate content, help improve writing and problem-solving, communicate, and engage in information searching. Regarding challenges, designers were primarily bothered by the low quality of ChatGPT-generated content, the limitations of ChatGPT itself, and their unpreparedness to embrace the tool.

Tools and Techniques

What are some of the tools and techniques that bring intrinsic value to the instructional design process or provide shortcuts that increase efficiency?

Brainstorming, Outlining, Mapping, Scripting

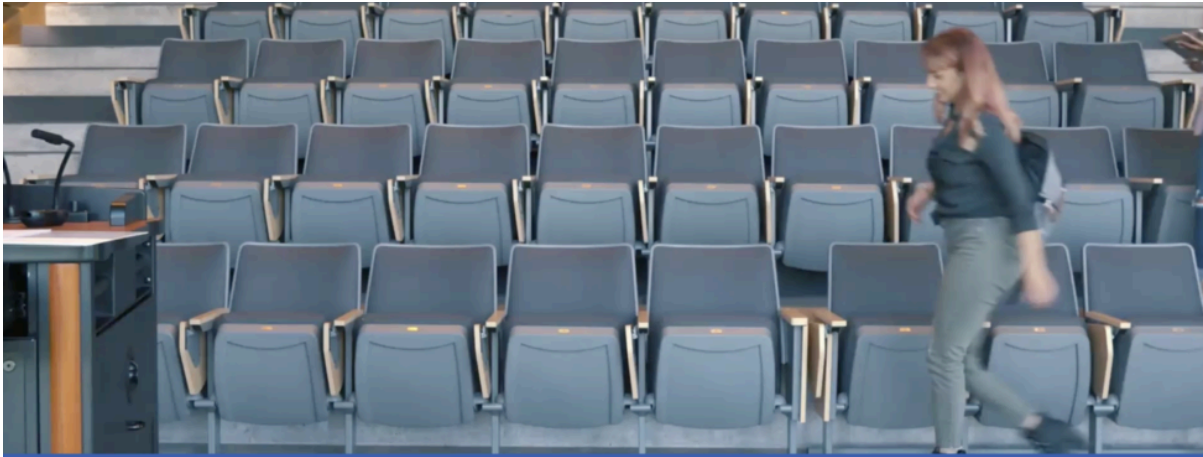
Generative AI accelerates instructional design, making it much quicker to move from rough ideas to ready-to-teach materials. Use cases are manifold. Here are a few ways I have used it:

- Act as a brainstorming partner, offering alternative ways to structure lesson plans and activities.
- Rewrite a syllabus or course map for a different modality or length of term
- Generate a variety of discussion and reflection questions based on lecture material or course readings
- Support alignment—especially when mapping activities and assessments to programmatic or professional standards
- Script a case study tied to learning objectives based on one or more news stories
- Draft learning objectives that match the intent and level of the activity.
- Estimate how long learners will take to complete activities.
- Create multiple-choice items, distractors and explanatory feedback

Multimedia: Visualizations and Audiovisual Content

AI-powered tools can convert text-based material into podcasts, videos, and infographics. AI-generated media assets are often impressive, though performance is seldom consistent. While the six-fingered hand is (mostly) a genAI artifact of the past, and newer models can now produce detailed prompts with increased accuracy, outtakes are plentiful and corrections often laborious. Personally, I use visualization tools such as [napkin.ai](#) and [NotebookLM](#) (infographics) frequently, and often create photorealistic images for case studies with various AI-tools such as [Adobe Firefly](#) or [Gemini Nano Banana](#). Occasionally, I use tools such as [Pictory](#), [invideo](#), [Copilot](#) or [NotebookLM](#) to turn a script or other source material into a narrated multimedia explainer video.

One of my favorite applications are case studies, where AI-generated avatars (produced with tools like [Heygen](#) or [Synthesia](#)) and voiceovers (e.g., [ElevenLabs](#)) can be used to narrate characters and offer digital storytelling that was previously out of budget (cf. Panke & Souders, 2025). However, there are limitations to the acting qualities of artificial intelligence characters – the average instructional design team does not have access to [Tilly Norwood](#).

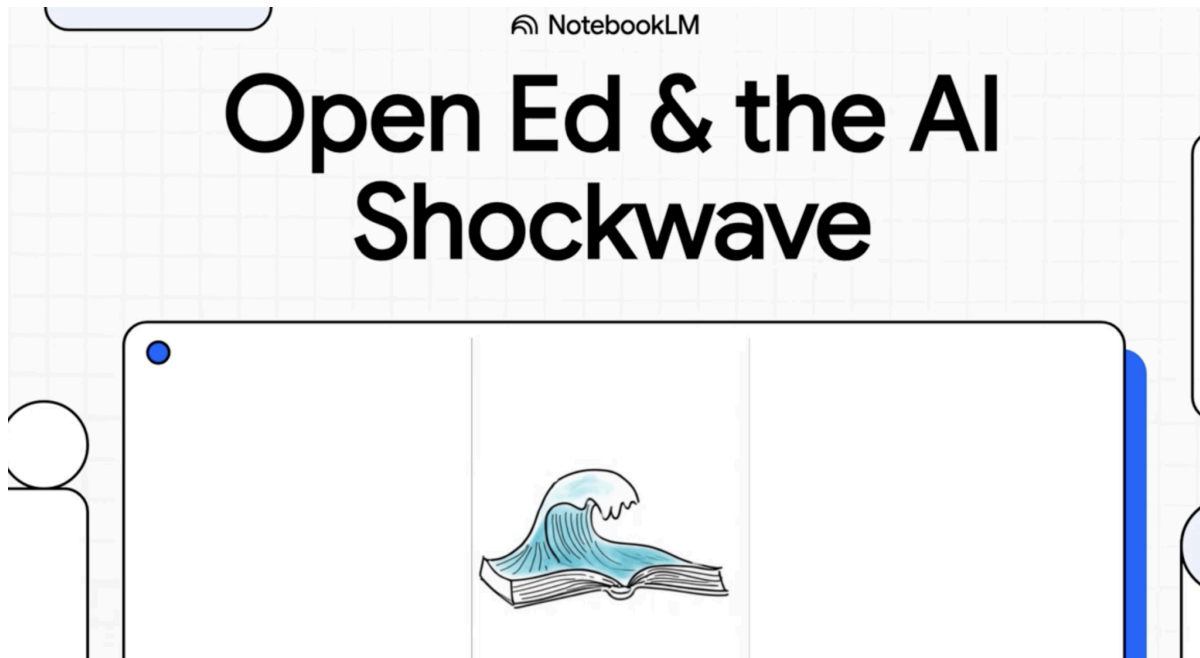


A university accessible to all abilities

Example 1: Copilot created explainer video on [Digital Accessibility](#) based on 2-page handout

A screenshot of a digital interface for an interactive case study. At the top, there is a small circular icon with a magnifying glass. Below it is a photograph of the U.S. Capitol building. The title "How a Bill Becomes a Law and Overcoming Setbacks" is centered in a large, white, sans-serif font. Below the title, a paragraph of text describes the activity: "In this activity, you will gain a comprehensive understanding of how a bill becomes a law in the U.S. Congress. You will walk through each step, from the committee stage to the final signature by the President. Along the way, you will encounter various decision-making moments that will challenge you to think critically and strategically." At the bottom, there is a green button with the text "Start the course" and a right-pointing arrow.

Example 2: Interactive case study of '[How a Bill Becomes Law](#)' (implemented with H5P, video material Heygen)



Example 3: [NotebookLM video](#) using an article as source article (Panke, 2024).

Digital Accessibility

Generative AI can support accessibility by helping to provide equivalent alternatives to visual and audiovisual content. AI can generate suitable alt-text descriptions for images and, more useful, generate long-form accessible descriptions for complex diagrams and charts, making complex visual content accessible to visually impaired learners. AI-powered summarization and transcription tools can convert videos into text (e.g., [youtube-transcript.io](#)) and significantly speed up the clean-up of automated transcripts,

Interactive Games

You can also use AI for rapid, prompt-based prototyping and interactive practice. For example, “vibe-coding” with Claude lets you describe an educational game in plain language (e.g., “Create a learning game about reading case law for social workers”) and iteratively refine the design by specifying things like style, layout, or colors—then share the finished game with learners. Beyond game creation, AI can serve as a dialogue partner for role-plays, responding in character based on a detailed script so students can practice interviewing, coaching, or de-escalation skills in a low-stakes setting.

Try it out: [Claude](#) Prompt: “Develop a learning game about...” (*you can specify style, colors, etc.; games are shareable*)



The Learning Designer

Advanced Cognitive Science Challenge

Your Mission

You've been hired as an instructional designer. Your job is to make evidence-based decisions about learning experiences based on cognitive science research. This expanded challenge tests your understanding of cognitive load theory, multimedia principles, generative learning strategies, memory and retention, and productive failure.

At the end, you'll receive a personalized learning profile with specific recommendations for deepening your understanding in areas where research challenges common intuitions.



Cognitive Load



Multimedia



Generative Learning



Memory & Retention



Productive Failure

Example 4: [The Learning Designer](#) – quiz on Instructional Design, generated with Claude

Instructional Design Futures

What will instructional design look like ten years from now? Fundamental change has happened to many sectors and workplaces. My mother, now in her eighties, worked in payroll accounting before I was born, and witnessed the transformation to electronic data processing as payroll cards and paper records were replaced by digital systems. The manual calculations of gross and net wages, tax deductions, and social insurance contributions were automated. Reports and statistics that previously had to be compiled laboriously were available at the push of a button. As a result, many payroll clerk jobs disappeared. The ones that remained shifted to data processing.

If the baseline becomes “anyone with ChatGPT can design a course,” institutions may deprofessionalize instructional design, treating it as a task rather than a discipline. What potentially remains resistant to automation is the more contextual work in leadership roles: understanding the political dynamics of a faculty committee, diagnosing when a learner’s

struggle signals a poorly designed assessment versus a gap in prerequisite knowledge, navigating the tension between conflicting stakeholder needs, making judgments about when to simplify complex material versus when complexity is the point.

Before electronic data processing, payroll clerks needed strong arithmetic skills. After automation, those skills became less critical. The computers did the calculations. What became more important were data entry accuracy and understanding how to operate the systems. Some of the most routine instructional design work may persist precisely because it requires the kind of precision and accountability that AI struggles with. Checking compliance with standards, ensuring that sensitive student data is handled appropriately, confirming that clinical or professional scenarios contain no factual errors —these tasks are often tedious, but carry consequences when done wrong.

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